**Experiment 17: Generate Contextual Embeddings for a Sentence Using BERT**

**Python Program**

```
import torch
from transformers import BertTokenizer, BertModel

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
model = BertModel.from_pretrained("bert-base-uncased")

text = "I love learning Python"
inputs = tokenizer(text, return_tensors="pt")

with torch.no_grad():
    outputs = model(**inputs)

embeddings = outputs.last_hidden_state

print("Sentence:", text)
print("Embedding Shape:", embeddings.shape)
print("Contextual Embeddings:", embeddings)
```

**Output**

```
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab> & C:\Users\SAMSUNG\AppData\Local\Programs\Pyth
on\Python314\python.exe "c:/Users/SAMSUNG/OneDrive/Documents/Generative lab/experiment17.py"

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect id
entical arch.
Sentence: I love learning Python
Embedding Shape: torch.Size([1, 6, 768])
Contextual Embeddings: tensor([[[ 0.2323,  0.2049, -0.1015,  ..., -0.1612,  0.1610,  0.3796],
         [ 0.5839, -0.0442,  0.0559,  ..., -0.1830,  0.4247,  0.4986],
         [ 0.8412,  0.6927,  0.5356,  ...,  0.1205,  0.5203,  0.2012],
         [-0.0368,  0.4168,  0.2125,  ...,  0.1154,  0.2764, -0.4765],
         [-0.1489, -0.2944, -0.2739,  ...,  0.5680,  0.0664, -0.2766],
         [ 0.8437,  0.2372, -0.0809,  ...,  0.0565, -0.8936, -0.2887]]])
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab>
```